CV of the researcher



• Personal information

Name: Rodríguez-Mayorga, Mauricio Antonio
Nationality: Spanish
ORCID: http://orcid.org/0000-0003-0869-9787
Research gate: https://www.researchgate.net/profile/Mauricio_Mayorga2
GitHub: https://github.com/marm314
Web site: https://marm314.github.io/index.html
email: marm3.14@gmail.com
h-index: 15, h10-index: 20

• Education

Jun. 26.06.2023 Master in Teaching in High-School, Baccalaureate, Vocational Training, and languages (**specialized in teaching Chemistry and Physics**), Univ. of Sevilla (Spain).

Jul. 24.07.2018 Ph.D. in Chemistry, Univ. Girona (Spain). Excellent Cum laude. Under the supervision of Prof. M. Solà and Dr. E. Matito

Sep. 06.09.2013 Master in Theoretical and Computational Chemistry, Univ. Rovira i Virgili (Spain). **(Graduated first in my class)**

Jul. 30.06.2012 BSc. in Chemistry, Univ. Granada (Spain). **(Graduated first in my class)**

• Employment history including current position/s **with the name/s of the advisor/s**

01.01.2026 **Postdoc researcher at Prof. P.-F. Loos and Dr. C. Mejuto-Zaera group,**
Laboratoire de Chimie et Physique Quantiques and Laboratoire de Physique
(current position) **Théorique - Toulouse (France)**

01.01.2025 - 31/12/2025 Postdoc researcher at Prof. P.-F. Loos group, Laboratoire de Chimie et Physique Quantiques - Toulouse (France)

01.07.2023 - 31/12/2024 Postdoc researcher at Profs. G. D'Avino and X. Blase group, Institute Néel Grenoble (France)

01.02.2023 - 31.06.2023 Postdoc researcher at Prof. J. C. Sancho-García's group, Universitat d'Alacant (Spain)

15.01.2021 - 31.12.2022 Postdoc researcher at Prof. L. Visscher's group, Vrjie Universiteit Amsterdam(Netherlands) with a Marie-Curie individual fellowship

01.01.2020 - 31.12.2020 Postdoc researcher at Dr. F. Bruneval's group, Univ. of Paris Saclay (France)

01.06.2019 - 30.11.2019 Postdoc researcher at Prof. S. Luber's group, Univ. of Zürich (Switzerland)

15.09.2013 - 24.07.2018 Ph.D. student, Univ. of Girona (Spain).

01.07.2012 - 06.09.2013 Research assistant, Univ. of Girona (Spain).

• Tutoring junior researchers (summary information, the names of the junior researchers should be indicated)

5) Mentoring: Final degree project of Mr. D. Këizer (2022). Center: Vrije Universiteit Amsterdam (Netherlands).

4) Mentoring: Final master project of Mr. C. Naim (2018). Center: Donostia International Physics Center (Spain)

3) Mentoring: Final master project of Mr. S. Sitkiewicz (2017). Center: University of the Basque Country (Spain)

2) Mentoring: Summer interships at DIPC for undergraduate students: Mr. Martí Gimferrer (2017), Mr. Carmelo Naim (2017), and Mr. Juan Reino (2016). Center: Donostia International Physics Center (Spain).

1) Mentoring: Final degree project of Ms. N. Abullí (2012). Center: University of Girona (Spain). This project was awarded with the extraordinary degree award.

• Teaching activities (summary information)

5) Subject: Physics and Chemistry to 2nd year High-school students. Center: I.E.S. Chaves Nogales (Sevilla). Credits: 10 ECTS, Dates: March 2023. [As part of the training of the master of education]

4) Subject: Integrated scientific techniques II. Center: University of Girona (Spain). Credits: 2.5 ECTS. Dates: January 2017.

3) Subject: Principles of chemical reactivity. Center: University of Girona (Spain). Credits: 4 ECTS. Dates: 26/09/16 to 30/09/2016

2) Subject: Principles of chemical reactivity. Center: University of Girona (Spain). Credits: 9 ECTS. Dates: 29/2/16 to 18/3/2016

1) Subject: Integrated scientific techniques II. Center: University of Girona (Spain). Credits: 5.5 ECTS. Dates: 10/11/2015 to 27/11/15

• Prizes, awards, fellowships

Jan. 2021 - Dec. 2022 ‘Marie Sklodowska-Curie individual fellowship’
Project: 891647-ReRDFMT

Aug. 2014 - Dec. 2017 ‘Beca de formación de Personal Universitario (FPU)’
(PhD grant, Spanish Goverment)

Feb. 2014 - Sep. 2014 'Beca de formació d'investigadors (FI)'
(PhD grant, Catalan government)- declined in favor to the FPU

Jul. 2013 Awarded with the Masters' Extraordinary Degree award from the University Rovira I Virgili (Spain).

Jul. 2012 Awarded with the Bachelors' Extraordinary Degree award from the University of Granada (Spain).

Jan. 2012 - Jun. 2012 'Beca Colaboración' (Highly competitive undergraduate fellowship form the Spanish Goverment for doing research at bachelor level)

• Personal skills (e.g. language skills, digital competences)

Spanish: native (mother tongue), English: fluent, and French: intermediate.

Good programming skills in Python, Fortran (F90) and C++.

Experience in Parallel programming (MPI and Open-MPI) and on building servers

• Most relevant publications

Papers signed as the corresponding author, paper signed as first author.

15) Rodríguez-Mayorga M., Loos P.-F., Bruneval F., Visscher L.; Time-reversal symmetry in RDMFT and pCCD with complex-valued orbitals **J. Chem. Phys.**, 162, 054716 (**2025**) doi.org/10.1063/5.0242504

14) Rodríguez-Mayorga M., Blase X., Duchemin I., D'Avino G.; From many-body ab initio to effective excitonic models: a versatile mapping approach including environmental embedding effects **J. Chem. Theory Comput.**, 20, 8675 (**2024**) doi.org/10.1021/acs.jctc.4c00756

13) Rodríguez-Mayorga M., Besalú-Sala P., Pérez-Jiménez A.J., Sancho-García J.C.; Application to nonlinear optical properties of the RSX-QIDH double-hybrid range-separated functional. **J. Comput. Chem.**, 1, 1 (**2024**) doi.org/10.1002/jcc.27302

12) Besalú-Sala, P., Bruneval, F., Pérez-Jiménez, Á. J., Sancho-García, J. C., Rodríguez-Mayorga, M.; RPA, an accurate and fast method for the computation of non-linear optical properties. **J. Chem. Theory Comput.** 19, 6062 **(2023)** doi.org/10.1021/acs.jctc.3c00674

11) Rodríguez-Mayorga M.; Standalone NOFT module (1.0). **Zenodo** (**2022**). doi.org/10.5281/zenodo.735273

10) Rodríguez-Mayorga M., Keizer D., Giesbertz K.J.H., Visscher L.; Relativistic effects on electronic pair densities: a perspective from the radial intracule and extracule probability densities. **J. Chem. Phys.**, 157, 194301 (**2022**) doi.org/10.1063/5.0127190

9) Rodríguez-Mayorga M., Giesbertz K.J.H., Visscher L.; Relativistic reduced density matrix functional theory **SciPost Chem.**, 1, 004 (**2022**) doi:10.21468/SciPostChem.1.2.004

8) Rodríguez-Mayorga M., Mitxelena I., Bruneval F., Piris M.; .Coupling Natural Orbital Functional Theory and Many-Body Perturbation Theory by Using Nondynamically Correlated Canonical Orbitals **J. Chem. Theory Comput.**, 17, 7562 (**2021**) doi.org/10.1021/acs.jctc.1c00858

7) Via-Nadal M., Rodríguez-Mayorga M., Ramos-Cordoba E., Matito E.; Singling Out Dynamic and Nondynamic Correlation. **J. Phys. Chem. Lett.**, 10, 4032 (**2019**) doi.org/10.1021/acs.jpclett.9b01376

Article awarded with the GEQC-RSEQ 2021 Methodology Development Prize

6) Rodríguez-Mayorga M., Ramos-Cordoba E., Lopez X., Solà M., Ugalde J. M., Matito E.; The Coulomb Hole of the Ne Atom. **Chemistry Open** ,8, 1 (**2019**) doi:10.1002/open.201800235 ***cover***

5) Rodríguez-Mayorga M., Via-Nadal M., Solà M., Ugalde J. M., Lopez X., Matito E.; Electron-Pair Distribution in Chemical Bond Formation. **J. Phys. Chem. A** 122, 1916 (**2018**) doi:10.1021/acs.jpca.7b12556

4) Rodríguez-Mayorga M., Ramos-Cordoba E., Via-Nadal M., Piris M., Matito E.; Comprehensive Benchmarking of Density Matrix Functional Approximations. **Phys. Chem. Chem. Phys.** 19, 24029 **(2017)** doi:10.1039/C7CP03349D

3) Mitxelena I., Piris M., Rodríguez-Mayorga M.; On the performance of Natural Orbital Functional Approximations in Hubbard Model. **J. Phys. Condens. Matter.** 29, 425602 **(2017)** doi:10.1088/1361-648X/aa80ca

Corrigendum: **J. Phys. Condens. Matter.** 30, 089501 **(2018)** doi:10.1088/1361-648X/aaa659

Article selected by the journal among the Highlights of the 2017

2) Rodríguez-Mayorga M., Ramos-Cordoba E., Feixas F., Matito E.; Electron Correlation Effects in Third-Order Densities. **Phys. Chem. Chem. Phys.** 19, 4522 (**2017**) doi:10.1039/C6CP07616E

1) Rodríguez-Mayorga M., Ramos-Cordoba E., Salvador P., Solà M., Matito. E.; Bonding description of the Harpoon Mechanism. **Mol. Phys.** 114, 1345 (**2016**) doi:10.1080/00268976.2015.1121297